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Tobacco smoking and sarcoidosis revisited—Evidence, mechanisms, and clinical implications: a narrative review

Nadia Sweis^{a,*}, Arnaud Dominati^{b,c,*}, Supriya Prasad^b, Fatima Alnaimat^d, Mina Alawqati^b, Israel Rubinstein^{e,f}, Rachel Caskey^g

^aDepartment of Business Administration, Princess Sumaya University for Technology, Amman, Jordan

^bDivision of Rheumatology, Department of Medicine, University of Illinois College of Medicine in Chicago, Chicago, IL, USA

^cDivision of Clinical Immunology and Allergology, Department of Medicine, Geneva University Hospital, Geneva, Switzerland

^dDivision of Rheumatology, Department of Internal Medicine, School of Medicine, University of Jordan, Amman, Jordan

^eDivision of Pulmonary, Critical Care, Sleep, and Allergy, Department of Medicine, University of Illinois College of Medicine in Chicago, Chicago, IL, USA

^fMedical and Research Services, Jesse Brown VA Medical Center, Chicago, IL, USA

^gDivision of Academic Internal Medicine, Department of Medicine, University of Illinois College of Medicine in Chicago, Chicago, IL, USA

Abstract

Sarcoidosis is a multisystem inflammatory disease with unclear etiology, influenced by genetic predisposition and environmental exposures. Smoking has been widely studied for its potential role in sarcoidosis, with conflicting evidence regarding its impact on disease risk, severity,

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CONTACT Nadia J. Sweis, N.Sweis@psut.edu.jo, World Health Organization Economic Consultant, World Bank Economic Consultant, Associate Professor of Economic, Department of Business Administration, Princess Sumaya University for Technology, P.O. Box 1438, Al-Jubaiha-Amman 11941, Jordan.

*Co-first authors: These authors contributed equally to this work.

Author contributions

Nadia Sweis: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Writing – original draft, Writing – review & editing; Arnaud Dominati: Conceptualization, Formal analysis, Funding acquisition, Methodology, Writing – original draft, Writing – review & editing; Supriya Prasad: Conceptualization, Formal analysis, Funding acquisition, Methodology, Writing – review & editing; Fatima Alnaimat: Conceptualization, Formal analysis, Methodology, Writing – review & editing; Mina Alawqati: Conceptualization, Formal analysis, Methodology, Writing – review & editing; Israel Rubinstein: Conceptualization, Formal analysis, Methodology, Writing – review & editing; Rachel Caskey: Conceptualization, Formal analysis, Methodology, Writing – review & editing.

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and treatment response. While some epidemiologic studies suggest that smoking is associated with a lower risk of sarcoidosis, others highlight variations based on geography, ethnicity, and smoking history. Assessing the effects of smoking is particularly challenging because of the complex composition of tobacco smoke, which contains thousands of chemicals with diverse biologic effects. Nicotine, a major component of tobacco, has demonstrated both pro- and anti-inflammatory properties, further complicating its role in sarcoidosis. This narrative review explored the complex relationship between smoking and sarcoidosis by examining smoking's effects on immune modulation, disease presentation and prognosis, and response to immunosuppressive therapy. By summarizing current evidence, this paper aimed to clarify the impact of smoking and nicotine on sarcoidosis and identify key areas for future research, particularly in understanding the mechanisms underlying smoking-related immune modulation and treatment outcomes.

Keywords

Cigarette; granulomatous disease; immune system; inflammation; inhalation; nicotine

Introduction

Sarcoidosis is a multisystem inflammatory disease characterized by the presence of noncaseating granulomas¹. The development of the disease is thought to be a result of the interplay between genetic predisposition—such as risk variants of human leukocyte antigen (HLA)—and external disease-promoting antigens^{1,2}. Gene-environment interaction studies have pinpointed loci that modulate sarcoidosis risk in the setting of exposures such as cigarette smoke or insecticides³. While the pathophysiology of sarcoidosis remains incompletely understood, advancements like single-cell RNA sequencing have implicated immune dysregulation in various cell types, including hyperactivity of CD4⁺ naïve T-cells and monocytes, dysfunction of regulatory T (Treg) cells, and anergy of effector T-cells⁴.

Sarcoidosis is a systemic disease that can affect multiple organs. Pulmonary involvement is observed in the majority of cases, while extrapulmonary manifestations occur with varying frequency. Ocular and cutaneous involvement are among the most common extrapulmonary presentations, whereas neurologic and cardiac manifestations are relatively less frequent.

Based on consensus criteria, sarcoidosis is diagnosed in the setting of compatible clinical and radiologic presentations, the presence of noncaseating granulomas on tissue biopsy, and the exclusion of other possible granulomatous disorders⁵. Of note, lupus pernio, Löfgren's syndrome (LS) (i.e. erythema nodosum, bilateral hilar lymphadenopathy, and polyarthralgia/polyarthritis), and Heerfordt's syndrome (i.e. parotid/salivary gland hypertrophy, facial nerve paralysis, and uveitis) are considered highly specific of sarcoidosis and thus may not require histologic confirmation for diagnosis in clinical practice⁶.

Epidemiologic studies suggest variations in sarcoidosis incidence and prevalence based on geographic region. East Asian countries such as South Korea, Taiwan, and Japan have an estimated prevalence of 1–5 per 100,000 individuals, compared with 140–160 per 100,000 in Sweden and Canada⁷. Similarly, the annual incidence of sarcoidosis is estimated to

be the least in East Asian countries at 0.51 per 100,000 and greatest in Scandinavian countries at 11–15 per 100,000. North American estimates generally fall between this range, at approximately 5–10 per 100,000. Race and ethnicity are believed to play a role, with African-Americans having a greater incidence in the United States than their white, Hispanic, and Asian American counterparts⁸. A few investigations suggest that sarcoidosis occurs more commonly in females, and other studies suggest no difference in incidence rates between females and males⁷. However, females tend to have a later peak age of diagnosis (50–65 years of age) versus males (20–45-years-old)^{7–10}. Of note, a recent study yielded greater incidence and prevalence rates for US military veterans compared with the general population, suggesting the role of veterans' unique exposure profiles in the development of sarcoidosis¹¹. Moreover, an increased incidence of sarcoidosis—25 cases per 100,000—was reported for firefighters involved in the US “9/11” rescue operations, underscoring the role of environmental exposures such as combustion byproducts in the disease's development¹².

A recent study examining the chronic respiratory disease burden from 1990 to 2019 across different regions and socioeconomic degrees reported that, smoking, air pollution, and occupational exposures were the top risk factors¹³. However, the relationship between tobacco smoking and sarcoidosis is a more complicated issue. The organs most commonly affected in sarcoidosis were those with the greatest environmental exposure, suggesting that factors such as particulate matter, microorganisms, and smoking may play a role¹⁴. In a study of 237 sarcoidosis patients with 474 matched controls, current daily smoking was associated with a lesser risk of sarcoidosis compared with ex-smoking. Moreover, the link between occupational exposures and sarcoidosis was stronger in never-smokers and individuals aged 40 years or older¹⁵.

Figure 1 provides a visual summary of the most relevant findings presented throughout this review, highlighting key insights from each section that explore the association between sarcoidosis and tobacco smoking or nicotine exposure.

Epidemiologic insights: smoking and sarcoidosis

Several global studies have suggested that sarcoidosis is less common in smokers. A US-based study of 706 cases and controls, matched by age, sex, and race, found a negative association between a history of ever smoking and sarcoidosis risk in a multivariable model (odds ratio [OR] 0.65, 95% CI 0.51–0.82, $p < 0.001$)¹⁶. Conversely, a study of US military veterans identified smoking as an independent risk factor for sarcoidosis, indicating a significant association with disease incidence (OR 1.36, 95% CI 1.31–1.41, $p < 0.0001$)¹¹. Given smoking is more intense among military veterans versus the general population¹⁷, these data suggest that the salutary effects of tobacco smoking in sarcoidosis express a bell-shaped dosage-response curve, and lower daily consumption is protective, while high daily tobacco use is deleterious.

A recent case-control investigation in Sweden revealed that in comparison with individuals without current nor past use of cigarettes, current smokers had lesser odds of having sarcoidosis (OR 0.48, 95% CI 0.32–0.71). The data also suggested that former smokers had increased odds of sarcoidosis, but this finding did not meet statistical significance (OR

1.33, 95% CI 0.98–1.81). Any history of smoking, however, was associated with lesser odds of sarcoidosis versus never smoking (OR 0.76, 95% CI 0.63–0.92)¹⁸. Nevertheless, the use of Swedish moist snuff, a smokeless tobacco that provides a high concentration of nicotine over prolonged periods, has not been indicated to affect the risk of sarcoidosis (relative risk [RR] 1.1, 95% CI 0.8–1.5), indicating that non-nicotine components of inhaled substances may influence disease risk¹⁹.

The relationship between smoking and sarcoidosis, however, is not unanimous. A retrospective study in Japan indicated that the prevalence of current smokers at the time of sarcoidosis diagnosis was greater than for the general Japanese population in all age groups other than men in their 30s, suggesting geography and ethnicity may impact smoking's relationship with sarcoidosis²⁰. A recent systematic review and meta-analysis involving seven case-control studies spanning multiple continents resulted in a negative association between sarcoidosis and smoking (random effect model OR 0.55, 95% CI 0.38–0.79)²¹. However, the authors described significant heterogeneity in the socioeconomic and demographic features of the different study populations, potentially introducing confounders. Four of the seven studies supported an inverse relationship between smoking and sarcoidosis^{22–25}. The remaining three studies in the systematic review, based in Sweden, the United States, and India, yielded no statistically significant correlations between cigarette smoking and the development of sarcoidosis^{21,26–28}.

Smoking and extrapulmonary manifestations

Several studies on the relationship between tobacco smoking and sarcoidosis focus on or involve predominantly pulmonary cases, raising the question of whether the possible protective effect of smoking may apply mainly to those with pulmonary involvement. A few groups of researchers have explored the association between smoking and certain types of non-PS. A case-control study of patients with cardiac sarcoidosis (CS) found a significant difference in the percentage of CS patients who ever smoked versus the percentage of controls who ever smoked (28.0% vs. 47.8%, $p = 0.0006$), suggesting a negative association between smoking and CS²⁹. Furthermore, when pre-treatment positron emission tomography (PET) scans were analyzed, individuals with a history of smoking were found to have had less inflammation in the myocardium compared with those who never smoked ($p < 0.0001$). This negative association between smoking and CS was also reported in another case-control study (18.6% vs. 47.2%) and it remained a significant association after multivariable analysis (OR 0.14, 95% CI 0.04–0.51, $p = 0.002$)³⁰.

Janot et al. conducted a retrospective chart review of 109 patients with biopsy-proven sarcoidosis. This study sample was further divided into patients with ocular sarcoidosis (OS) versus those without. The OS group had a greater percentage of smokers than the non-OS group (71.4% vs. 42.0%, $p = 0.027$), suggesting a positive association that contrasts with several other studies of smoking and sarcoidosis. In a multivariate regression analysis, ever-smoking remained a significant independent risk factor for OS (OR 5.25, 95% CI 1.58–17.41, $p = 0.007$)³¹. Another study, which followed 518 predominantly African-American patients over five years, reported an 11-fold increased risk of developing extrapulmonary complications (95% CI 5.2–11.1, $p < 0.001$) for smokers, with uveitis being the most

common³². A group in the Netherlands conducted a prospective cohort study involving patients with sarcoid arthritis. Results found a negative association between sarcoid arthritis and smoking (OR 0.09, 95% CI 0.02–0.37)³³. Table 1 summarizes studies investigating the association between smoking and sarcoidosis.

Cardiovascular risk factors and sarcoidosis

In a US study of 803,576 sarcoidosis patients—of whom 4,177 experienced sudden cardiac arrest—traditional cardiac risk factors such as hypertension, hyperlipidemia, and smoking were significantly lesser for those with sudden cardiac arrest ($p < 0.001$)³⁴. Notably, even after multivariate analysis, smoking remained significantly less for sarcoidosis patients with sudden cardiac arrest (OR 0.83, 95% CI 0.69–0.99, $p = 0.046$).

Tobacco smoking-related immune modulation

Tobacco smoke is a complex aerosol comprising nicotine, a parasympathetic alkaloid that functions as a cholinergic agent, along with approximately 600 other distinct ingredients. When combusted, these substances generate more than 7,000 chemical compounds, many of which are toxic or carcinogenic³⁵. This extensive chemical diversity makes it challenging to elucidate the molecular mechanisms underlying the immune cell alterations induced by smoking, a task further complicated by conflicting experimental results. To better understand these effects, nicotine has been studied separately for its potential immune-modulating properties. Moreover, the impact of both smoking and nicotine is shaped by dosage, method of administration, and the specific cellular, organ, and disease context in which they exert their effects^{36–38}.

Experimental evidence highlights the conflicting effects of nicotine versus cigarette smoking on rheumatoid arthritis (RA). In rodent models, nicotine administration significantly reduces key pro-inflammatory cytokines and alleviates clinical symptoms^{39,40}. In stark contrast, epidemiologic data consistently link cigarette smoking to an increased risk of RA, with smoking exacerbating the condition through interactions with genetic risk factors such as HLA-DRB1⁴¹. Similarly, while smoking appears to predispose individuals to multiple sclerosis, experimental studies using an animal model have demonstrated that nicotine alone can control the disease⁴². Moreover, when nicotine was removed from cigarette smoke, the remaining compounds exacerbated the condition, suggesting that other constituents of cigarette smoke contribute to its deleterious effects. Together, these findings highlight the complex and sometimes opposing roles of nicotine and the other compounds present in cigarette smoke in modulating immune responses.

Covering the nicotine dosage-dependent effects, Nordman et al. found that mice treated with a 1- μ M nicotine exhibited a marked increase in anti-inflammatory cytokines IL-4 and IL-10 concentrations, along with a significant decrease in IFN- γ , a pro-inflammatory cytokine⁴³. By contrast, greater dosages of nicotine induced an exaggerated Th2 polarization, which was associated with an increased incidence of allergic and autoimmune diseases. Moreover, Kikuchi et al. suggested that long-term nicotine treatment exacerbates Crohn's disease by increasing Th1 concentrations⁴⁴. In mouse models of Crohn's disease, the impact of nicotine was dosage-dependent and inflammation site specific (e.g. colon vs. small intestine). These

animal studies underscore the importance of nicotine dosage, treatment duration, and the context in which it is administered in modulating immune responses. On the other hand, smoking has been linked to an increased risk of Crohn's disease while being associated with a decreased risk of ulcerative colitis¹⁹.

Immune-modulating effects of smoking over time

In a time-dependent investigation, Saint-André et al. examined how smoking influences the immune system in healthy volunteers and demonstrated that while the pro-inflammatory effects of smoking on the innate immune system are rapidly reversible upon cessation, the alterations in adaptive immune responses endure for years³⁸. These enduring changes are thought to be mediated by epigenetic memory, involving long-lived B- and T-cell subsets and DNA methylation changes that affect signal trans-activators and metabolic regulators. Furthermore, smoking-related variables, including the duration of smoking, the time since quitting, and the total number of cigarettes smoked, consistently correlated with cytokine production³⁸.

Immune-modulating effects of nicotine

The immune-modulating effects of nicotine are mediated through the nicotinic acetylcholine receptor $\alpha 7$ ($\alpha 7nAChR$), which is essential for acetylcholine's role in fostering an anti-inflammatory environment. Activation of this receptor enhances Treg cell responses, down-regulates pro-inflammatory cytokine production, and promotes apoptosis of pro-inflammatory cells^{45–47}. In particular, nicotine has been demonstrated to inhibit the release of tumor necrosis factor (TNF) from macrophages⁴⁸, while also suppressing the NF- κ B pathway and reducing the secretion of other pro-inflammatory cytokines³⁷.

Various studies have reported a positive impact of nicotine across different diseases, such as inflammatory bowel disease⁴⁹, sarcoidosis^{50,51}, hypersensitivity pneumonitis^{52,53}, mucocutaneous involvement in Behçet's disease⁴⁵, and multiple sclerosis in animal models⁴⁵. However, the protective effects of nicotine were typically short-term and have not been evaluated in long-term scenarios.

Smoking and genetic factors in sarcoidosis

A Swedish study investigated the interplay between gene polymorphism and smoking in sarcoidosis, revealing that smoking alters the risk of the disease by 56% in patients with LS and 62% in patients with non-LS²⁶. Specifically, the interleukin-23 receptor (IL-23R) gene variant rs61780312 emerged as a risk locus for non-LS sarcoidosis among smokers, with an OR of 3.23 (95% CI 1.75–5.97). The IL-23R is a pro-inflammatory cytokine receptor that stimulates Th17 cells and other cytokines involved in cell-mediated inflammatory processes. Similarly, the Fc receptor-like 1 (FCRL1) gene variant rs12132140 was identified as a risk factor for LS in smokers, yielding an OR of 2.92 (95% CI 1.68–5.07). The FCRL1 is a membrane protein of the immunoglobulin superfamily receptor that is predominantly expressed by B-cells²⁶.

Smoking and risk of infection

Cigarette smoke damages the respiratory epithelium by directly impairing the cells that maintain the epithelial barrier, affecting both cell replenishment and ciliary motion. It also reduces cough reflex sensitivity, further hindering pathogen elimination³⁵. Smoking's impact on immunity is complex, exhibiting both pro-inflammatory and immunosuppressive effects. In the innate immune system, short-term exposure activates inflammatory cells. However, long-term exposure impairs the phagocytic functions of macrophages and neutrophils, disrupts dendritic cell activity, and reduces natural killer cell function, thereby weakening host defenses. In adaptive immunity, smoking is linked to lymphopenia, reduced B- and T-cell responses, and lower concentrations of immunoglobulins and cytokines, although some immune cell counts may paradoxically increase³⁵. Collectively, these changes contribute to an elevated risk of respiratory infections in a dosage-dependent manner, while smoking cessation can help restore immune function and reduce infection-related morbidity³⁵.

Impact of tobacco smoking on clinical features and management of sarcoidosis

Several retrospective studies have investigated the relationship between smoking status and the clinical outcomes of pulmonary sarcoidosis (PS), yielding mixed results.

Bronchoalveolar lavage fluid

Bronchoalveolar lavage fluid (BALF) studies of patients with PS have examined several cellular and cytokine variables, revealing distinct differences between smokers and non-smokers. For context, even in individuals without sarcoidosis, Forsslund et al. demonstrated that active smoking increases the percentage of CD8⁺ T cells while reducing the percentage of CD4⁺ T-cells in both BALF and peripheral blood⁵⁴. In sarcoidosis, regarding total cell counts, multiple investigations consistently reported greater cell counts in smokers compared with non-smokers^{24,55–60}. In parallel, the percentage and absolute numbers of alveolar macrophages were generally elevated in smokers, suggesting a smoking-related accumulation of these cells in the lung^{24,55,56,58,60–62}. Conversely, the lymphocyte compartment had a significant reduction in lymphocyte percentages in smokers in most of the studies^{24,55–60,63}. Similarly, alterations in T-lymphocyte subpopulations have been observed, with several studies noting significantly lesser CD4/CD8 ratios for smokers^{24,55,57,64}, corroborating findings from Forsslund et al. discussed above⁵⁴. However, some did not find statistically significant differences^{65,66}. Neutrophil percentages appear not to be affected, with only one out of five studies reporting significant differences between smokers and non-smokers⁶⁵.

From lipopolysaccharide (LPS)-stimulated alveolar macrophages, Yamaguchi et al. reported a significant decrease in the release of IL-1/IL-1 β and TNF in smokers^{55,56} while another study did not⁶⁷. Moreover, Urbankowski et al. reported no significant differences in median concentration of IL-6, IFN- γ , and IL-17A between non-smokers and smokers. In addition, in a study involving BALF from 57 PS patients, a significant difference was noted in the concentration of soluble Fas, an inhibitor of apoptosis, between smokers and non-smokers

($p = 0.03$), with an observed correlation between soluble Fas concentrations and lymphocyte percentages ($r = 0.36$, $p < 0.05$)⁶⁸. This finding suggests a possible role for lower soluble Fas concentrations in promoting the resolution of granulomas by increasing the apoptosis rate of inflammatory cells.

Overall, these BALF studies indicate that smoking in PS is associated with an altered cellular profile—characterized by increased total cell counts and alveolar macrophages, along with reduction in lymphocyte percentages, CD4/CD8 ratio, along with variable effects on cytokine release. This reflects the complex immunologic impact of smoking in this condition. Notably, an excessive number of alveolar macrophages can inhibit the proliferation of T lymphocytes⁶⁹, further supporting the aforementioned findings and the observed lesser incidence of PS for smokers. Table 2 summarizes BALF studies of patients with PS.

Pulmonary function test

Pulmonary function tests (PFTs) have been extensively used to assess the impact of smoking on lung function in patients with PS. Across multiple studies, several key parameters have been evaluated, including total lung capacity (TLC), forced expiratory volume in one second (FEV₁), forced vital capacity (FVC), the FEV₁/FVC ratio, and diffusion capacity for carbon monoxide (DL_{CO}).

For TLC, most investigations reported comparable values between smokers and non-smokers^{24,32,60,70}. In contrast, Hanley et al. documented significantly lesser TLC values for smokers ($72.7 \pm 3.5\%$ vs. $82.5 \pm 3.3\%$ for non-smokers, $p < 0.05$), along with a trend toward a decline in all pulmonary function parameters in another study⁶⁴.

The results for FEV₁ were more variable. While two studies reported no significant differences between smokers and non-smokers^{70,71}, four others found that smokers had significantly lesser FEV₁ values^{32,60,68,72}, underscoring a trend toward diminished airflow in this group. FVC values followed a similar pattern. Even in studies in which the difference did not reach statistical significance, there was a clear trend toward reduced FVC for smokers in most of them (Table 3). For FEV₁/FVC ratio, most studies reported no significant differences between the two groups^{24,55,70,71,73}. However, two other studies documented a significantly lesser FEV₁/FVC ratio for smokers, underscoring the well-known negative impact on airway obstruction^{32,74}.

DL_{CO} also yielded mixed results from two of seven studies yielding statistical significance^{32,55}. Although Boros et al. noted statistically significant reductions in both lung volumes and DL_{CO} in ever-smokers, differences were considered of questionable clinical importance⁷⁵. Moreover, in a Turkish study of 257 patients with PS, no correlation was observed between spirometric findings, including DL_{CO} and smoking status⁶⁵.

These overall findings should be interpreted with caution, given the inherent heterogeneity between studies. Differences in disease staging, racial backgrounds, pack-year exposures, and duration of smoking, as well as variations in the classification of smokers (current versus ever-smokers) versus never-smokers, all contributed to the variability in reported

outcomes. The variability in PFT results complicates the assessment of whether observed lung function impairments in smokers are a result of COPD, the exacerbation of sarcoidosis-related inflammation, or a combination of both.

Imaging studies

Imaging studies comparing (ever-)smoking and non-smoking patients with PS reveal several noteworthy differences. In an early investigation by Valeyre et al. chest radiographs revealed no significant differences between smokers and non-smokers at initial evaluations and during follow up²⁴. Another study reported that chest radiographic parenchymal abnormalities were more common in ever-smokers than in non-smokers (59% vs. 46%, $p = 0.026$)⁷². A Japanese study found that emphysema was markedly more prevalent in smokers compared with non-smokers (48% vs. 4%, $p = 0.002$), despite no significant differences in other parenchymal findings, such as consolidation, ground-glass attenuation, reticular opacities, and small nodules⁷¹. Similarly, Gerke et al. observed that smokers had a greater prevalence of emphysema (18% vs. 1%, $p = 0.001$) but a significantly lesser frequency of bronchoalveolar bundle thickening (29% vs. 58%, $p = 0.003$) compared with non-smokers⁷⁰. This finding may be explained by reduced inflammatory infiltrates in smokers. Furthermore, a review of 150 ever-smoking patients with PS described that 33% of ever-smokers exhibited severe fibrosis combined with emphysematous and bullous changes on CT scans, a pattern distinct from the typical upper-lobe predominant emphysema observed in pure smoking-related disease³². Moreover, the authors found that most of these patients were younger and had relatively lesser pack-year smoking histories compared with reported pack-years leading to COPD in other studies.

Adding further evidence, a Greek cohort of 87 PS patients with increased lung 18 F-FDG uptake demonstrated that ever-smokers had significantly lesser $DL_{CO}\%$ -predicted values and greater total lesion glycolysis versus never-smokers ($p = 0.032$ and $p = 0.005$, respectively)⁷⁶. Together, these findings indicate that, while overall radiographic staging may appear similar between smokers and non-smokers, smoking is associated with greater prevalences of emphysema, distinctive fibrotic changes, and alterations in lung function and metabolic activity, underscoring smoking's impact on the pulmonary manifestations of sarcoidosis.

Prognosis (chronicity, infection, relapse, and O₂ therapy)

Several studies have investigated the impact of smoking on the prognosis of sarcoidosis, examining factors such as disease chronicity, relapse, and the need for oxygen therapy. For example, a large Serbian study of 2,380 patients found no significant differences in smoking status between those with acute and chronic forms of the disease⁷⁷. Another study involving 306 patients revealed no difference in spontaneous remission rates between smokers and non-smokers with PS⁷⁸. Moreover, smoking did not influence pulmonary disease progression, functional or radiologic, at a 5-year follow up of patients with sarcoidosis in a European study⁷⁹. In addition, a Turkish study of 275 sarcoidosis patients, including 263 with a follow-up period exceeding 24 months, found no significant difference between smokers and non-smokers in disease progression or improvement⁸⁰.

In contrast, research in a cohort of 96 Chinese patients with PS undergoing corticosteroid treatment demonstrated that a history of smoking was significantly associated with an increased risk of relapse (hazard ratio [HR] 3.674, 95% CI 1.573–8.581, $p = 0.003$)⁸¹. Further, in a study of 28 sarcoidosis patients, ex-smokers began long-term domiciliary oxygen therapy at a significantly younger age than never-smokers, irrespective of chronic airway disease⁸². Another study, which followed 518 predominantly African-American patients for 5 years, reported that 10.4% required home oxygen therapy, with 78% of these patients being former smokers³². Conversely, a separate cohort of 60 patients revealed that smoking status did not significantly affect the extent of disease, the frequency of extrathoracic involvement, or corticosteroid usage⁸³. This observation aligns with findings that smoking can exacerbate disease severity and relapse risk, further highlighting the complex role of smoking in sarcoidosis progression.

For infection risk, one study examined the frequency of acute worsening events in PS, which is defined as episodes lasting 4 weeks or fewer that require antibiotics or increased corticosteroid dosages. No significant relationship between smoking history and the frequency of these acute events was observed⁸⁴.

Of patients with LS, Grunewald et al. found that they were generally younger, previously healthy, and more likely to have been non-smokers compared with non-LS patients. They also tended to have more favorable prognoses, which may be partially influenced by their non-smoking statuses⁸⁵.

These findings suggest complex interactions between smoking and sarcoidosis, affecting various clinical aspects differently across studies.

Limitations

The conflicting epidemiologic data and clinical observations on the relationship between tobacco smoking and sarcoidosis likely stem from a combination of methodologic and biologic factors. Many of the available studies were retrospective, which may have introduced selection and recall biases. In addition, substantial heterogeneity existed between study populations, including differences in ethnicity and geographic origin, which may reflect underlying genetic predispositions to sarcoidosis. Smoking status is also inconsistently classified. Some studies analyzed ever-smokers separately from current smokers, while others combined these groups, thus making comparisons challenging. Further confounding factors included sex, smoking intensity and duration—as observed in cohorts of US veterans—as well as occupational exposures. Together, these variables complicate interpretation of the existing literature and highlight the need for harmonized, prospective studies to better clarify the role of tobacco smoking in sarcoidosis.

However, given the relative rarity of sarcoidosis, conducting large-scale randomized controlled trials remains logistically challenging. As a result, much of the available evidence we have is derived from observational and case-control studies, which, while inherently limited, offer valuable insights. Real-world data from diverse populations—such as US veterans, Swedish registries, and other international cohorts—provide important information, particularly in the consideration of the heterogeneous manifestations of

sarcoidosis (e.g. pulmonary, cardiac, ocular involvement). In this context, observational studies represent the most practical and scientifically valid strategy for guiding clinical understanding and informing hypotheses for future prospective research.

Transdermal nicotine treatment in sarcoidosis

Despite extensive preclinical and epidemiologic data suggesting potential benefits of nicotine in sarcoidosis, only two randomized controlled trials have been conducted, both by the same research group^{50,51}. These studies, carried out in the United States in 2013 and 2021, evaluated transdermal nicotine patch therapy in symptomatic patients with active PS. The earlier trial was an open-label study of 13 participants⁵⁰, while the later trial was a double-blind study of 49 patients⁵¹. Nicotine treatment followed a stepwise dosage-increase regimen to minimize adverse effects, with increases from 7 mg to 14 mg to 21 mg at weekly intervals, as tolerated. The treatment duration was 12 weeks in the earlier study and 24 weeks in the later one. Both trials reported that nicotine was well-tolerated, with no serious adverse events. Minor adverse effects—such as headaches, dizziness, insomnia, agitation, skin irritation, and abnormal dreams—were most common during dosage increases but typically resolved over time. Importantly, neither study observed nicotine dependency in participants.

The trials yielded mixed results for both lung function and symptom improvement. In the 24-week study, nicotine-treated patients experienced a statistically significant 140-mL increase in FVC (95% CI 10–260, $p = 0.015$), whereas the 12-week study did not yield any significant changes in FVC. In addition, neither study resulted in meaningful improvements in sarcoidosis-related symptoms, as assessed by the St. George's Respiratory Questionnaire.

Beyond lung function, nicotine's immunomodulatory effects were investigated in the earlier study, which included 19 participants stratified into symptomatic and asymptomatic groups. The researchers found that symptomatic patients exhibited impaired peripheral immune responses to TLR2, TLR4, and TLR9 ligands, along with reduced Treg cell populations. Following 12 weeks of transdermal nicotine therapy, a subset of symptomatic patients experienced restored TLR2 and TLR9 responsiveness and an increase in Treg cells⁵⁰.

While nicotine demonstrates anti-inflammatory effects, its toxicity and addictive potential limit its clinical applicability. To harness its therapeutic benefits while minimizing risks, future research should focus on identifying selective $\alpha 7$ nAChR agonists with improved safety profiles. Moreover, no studies have explored nicotine therapy beyond 24 weeks in sarcoidosis, and its effects likely depend on multiple factors, including dosage, treatment duration, and mode of administration (oral, inhaled, injected, or transdermal). Larger, long-term trials are necessary to better define its role in PS management.

$\alpha 7$ nAChR agonists: a therapeutic avenue in sarcoidosis?

Beyond nicotine itself, several research groups have investigated selective agonists of $\alpha 7$ nAChR as potential therapeutic agents in autoimmune and inflammatory diseases, as summarized in recent comprehensive reviews^{86,87}. In rheumatoid arthritis preclinical studies, $\alpha 7$ nAChR agonists such as GTS-21 (a partial agonist), PNU-282987 ($\alpha 7$ -selective),

and Sinomenine (a plant-derived alkaloid) have demonstrated anti-inflammatory effects, including reduced joint inflammation and cytokine production^{86,88,89}. In addition, a recent study using PBMCs from rheumatoid arthritis patients demonstrated that GTS-21 suppressed Th17 differentiation while also promoting Th₂ polarization *via* $\alpha 7$ nAChR signaling, which reinforced the immunomodulatory role of this receptor⁹⁰.

In models of inflammatory bowel disease, $\alpha 7$ nAChR activation has similarly been associated with reduced mucosal inflammation and enhanced intestinal barrier integrity^{91,92}. These effects are thought to be mediated through the cholinergic anti-inflammatory pathway, dampening NF- κ B signaling in macrophages and intestinal epithelial cells.

In psoriasis, increased expression of $\alpha 7$ nAChR has been observed in lesional skin, particularly in keratinocytes and macrophages. Treatment with the $\alpha 7$ -selective agonist PNU-282987 alleviated psoriasis-like inflammation in mice by suppressing STAT3 and NF- κ B signaling, whereas antagonizing $\alpha 7$ nAChR worsened inflammation⁹³. PNU-282987 treatment alleviated psoriasis-like inflammation by down-regulating the expression of multiple types of pro-inflammatory mediators through inhibition of the STAT3 and NF- κ B signaling pathways, whereas $\alpha 7$ nAChR antagonist treatment exacerbated its effect.

While a wealth of preclinical research has demonstrated the potential value of $\alpha 7$ nAChR-targeting agents in various inflammatory disease models, clinical translation data remain limited. These findings support the hypothesis that nicotine's observed effects in sarcoidosis may be mediated, at least in part, through activation of this receptor. Notably, besides nicotine, no clinical trials have evaluated $\alpha 7$ nAChR agonists in sarcoidosis—or in other inflammatory or autoimmune diseases—highlighting a promising and underexplored avenue for future therapeutic development.

Immunosuppressant treatment and tobacco smoking

While no studies have evaluated the impact of smoking on immunosuppressant therapy in sarcoidosis, research in other areas provides some insight. Several studies have examined the efficacy of corticosteroids in smokers, particularly focusing on inhaled corticosteroids (ICS) in COPD patients. A systematic review found that heavier or current smokers did not experience the same benefits from ICS as lighter or ex-smokers, as measured by lung function and exacerbation rates. However, the clinical importance of these effects was questioned⁹⁴. Further research involving oral corticosteroids indicated varied results. In a randomized, placebo-controlled crossover study, patients with asthma received either oral prednisolone (40 mg daily) or placebo for 2 weeks. Never-smokers had significant improvements in FEV₁, morning peak expiratory flow, and asthma control scores following prednisolone treatment, whereas smokers did not experience significant changes⁹⁵.

A systematic review and meta-analysis of patients with rheumatoid arthritis assessed responses to methotrexate (MTX). The findings suggested that smokers were 58% more likely to have had an inadequate response to MTX compared with non-smokers (OR 1.58, 95% CI 1.21–2.06, $p = 0.001$, $I^2 = 69.3\%$)⁹⁶. In addition, a retrospective study on MTX treatment for ectopic pregnancies in 299 patients found that smoking was independently

associated with treatment failure, with an OR of 2.58 (95% CI 1.16–5.75)⁹⁷. MTX acts mechanistically by being converted into polyglutamylated forms (MTX-PG), which are retained inside immune cells and inhibit key enzymes involved in inflammation. Adenosine triphosphate-binding cassette (ABC) transporters, a protein family responsible for drug efflux, can extrude MTX from cells, reducing MTX's efficacy. Smoking appears to upregulate these transporters, potentially leading to lower intracellular retention of MTX, thus diminishing its anti-inflammatory effects⁹⁸. These findings indicate that smoking may adversely affect the efficacy of corticosteroids and MTX, highlighting the potential for smoking to influence the effectiveness of immunosuppressants in other conditions, including potentially sarcoidosis.

Public policy, economic burden, and evolving perspectives

The economic burden of smoking-related diseases is well-documented in public health research, yet its implications for sarcoidosis remain largely unexplored. While extensive data exist on the economic impact of smoking cessation in conditions like lung cancer and cardiovascular disease, the relationship between smoking and sarcoidosis is less established.

From a health economics perspective, the price elasticity of tobacco use is well-known across diverse populations⁹⁹. Indeed, studies have demonstrated that significant increases in cigarette prices—often through taxation—can reduce smoking prevalence, particularly for men. However, these economic models have not been examined in the context of sarcoidosis. In the absence of a clear causal link between smoking and disease progression in sarcoidosis, policymakers face challenges in designing targeted and effective interventions.

Conclusions

The relationship between smoking and sarcoidosis remains complex and, at times, paradoxical. Some epidemiologic studies strongly suggest that current smoking may reduce sarcoidosis risk, while others emphasize smoking's detrimental effects on lung health. Conflicting evidence indicates that nicotine's immunomodulatory properties could influence disease activity, complicating public health messaging, especially since standard anti-smoking campaigns may not fully address sarcoidosis-specific nuances.

Research, particularly longitudinal studies and randomized controlled trials is warranted to elucidate the impact of smoking cessation on sarcoidosis progression, health care costs, and patient outcomes. Collaboration among epidemiologists, policymakers, and basic scientists is crucial for developing evidence-based strategies. Exploring nicotine's potential role as an adjunct therapy in certain contexts warrants investigation. Finally, integrating health economics into sarcoidosis research could inform cost-effective prevention and treatment, highlighting the need for a balanced approach that reconciles traditional tobacco control measures with emerging insights into nicotine's effects in sarcoidosis.

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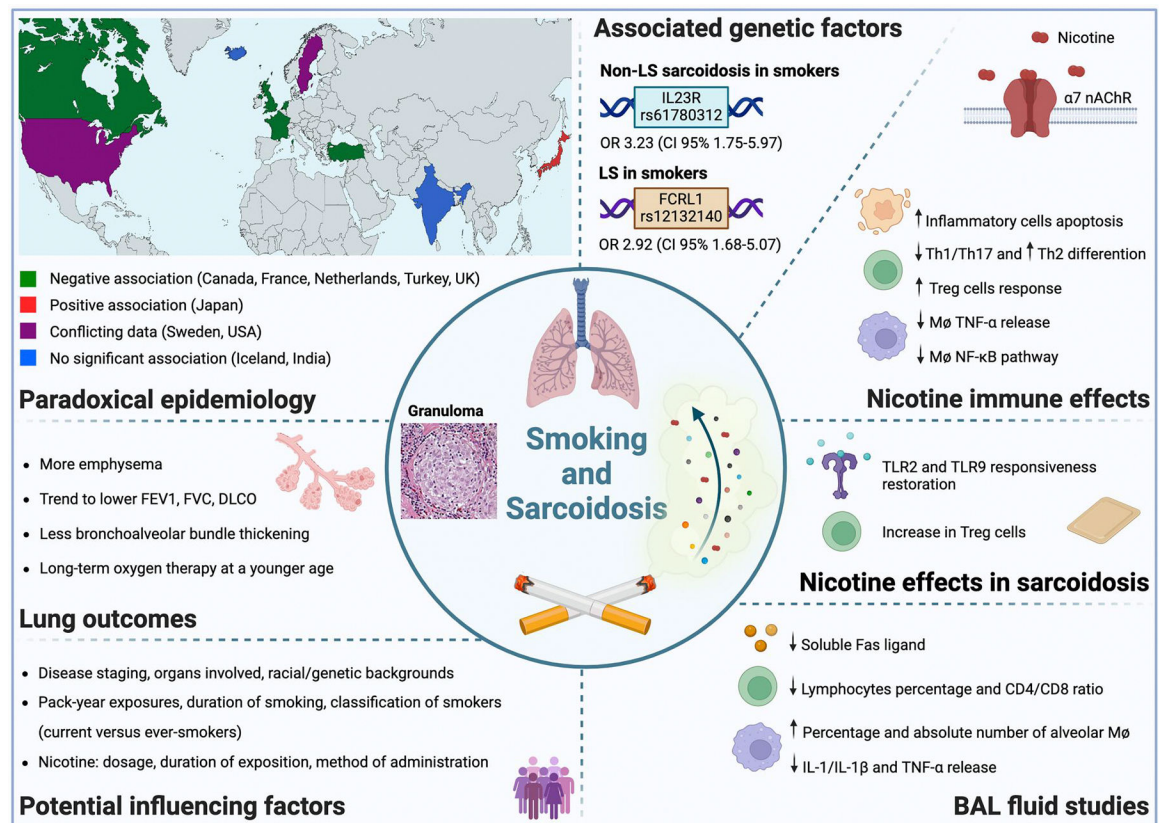


Figure 1.

Summary/Central figure of association between sarcoidosis and tobacco smoking/nicotine (created in <https://BioRender.com>). α7 nAChR: alpha-7 nicotinic acetylcholine receptor; BAL: bronchoalveolar lavage; CI: confidence interval; DL_{CO}: diffusing capacity of the lungs for carbon monoxide; FCRL1: Fc receptor-like 1; FEV₁: forced expiratory volume in one second; FVC: forced vital capacity; IL23R: interleukin-23 receptor; LS: Lofgren Syndrome; Mφ: macrophages; NF-κB: nuclear factor kappa-light-chain-enhancer of activated B-cells; OR: odds ratio; Th: helper T-cells; TLR: toll-like receptor; TNF: tumor necrosis factor; Treg: regulatory T-cell.

Table 1.

Summary of studies investigating the association between smoking and sarcoidosis.

References	Country	Case-control study	Number of patients with smoking history Sarcoidosis/controls	Female	NS/S or NS/FS/S or NS/ES ^a M/F or Total (%)	Control group	Adjusted Odds Ratio (95% CI) or Relative Risk (95% CI) ^b	Association between smoking and sarcoidosis
Pulmonary sarcoidosis								
Bresnitz et al. 1986 ²⁷	USA	Yes	50/105	10%/11%	44/28/28	39/20/41	0.72 (0.32–1.72) ^c	No significant association
Newman et al. 2004 ¹⁶	USA	Yes	706/706	64%	55/45 ^a	43/57	0.65 (0.51–0.82)	Negative association
Ungprasert et al. 2016 ²²	USA	Yes	345/345	50%	60/21/19	42/22/36	0.48 (0.35–0.66)	Negative association
Seedahmed et al. 2023 ¹¹	USA	Yes	23,747/71,241 (US veterans)	13.5%	26/74 ^a	36/64	1.36 (1.31–1.41)	Positive association in US veteran population
Carlens et al. 2010 ¹⁹	Sweden	No	342	0%	ES (never SMS): 27% Ever SMS (NS); 18%	National survey of sarcoidosis incidence in males	Tobacco smoking: 0.5 (0.4–0.6) ^b SMS: 1.1 (0.8–1.5) ^b	Negative association
Rivera et al. 2019 ²⁶	Sweden	Yes	747/2,966	43%/74%	52/48 ^a	45/55	LS: 0.84 (0.64–1.09) non-LS: 0.92 (0.74–1.13)	No significant association
Dehara et al. 2023 ¹⁸	Sweden	Yes	165/660	38%	63/27/9	53/25/21	NS vs. S: 0.48 (0.32–0.71) NS vs. ES: 0.76 (0.63–0.92)	Negative association
Harf et al. 1986 ²⁵	France	Yes	101/404	48%	75/25	44/56	0.44 (0.28–0.71)	Negative association
Valeyre et al. 1988 ²⁴	France	No	64		70/30	National survey 64/46	NA	Lower prevalence of smoking in patients with sarcoidosis than general population (p<0.05).
Douglas et al. 1986 ²³	United Kingdom	No	183	51%	78/22	National survey	0.36 (0.25–0.52) ^b	Negative association
Gupta et al. 2010 ²⁸	India	Yes	98/196	37%	88/12	85/15	0.75 (0.35–1.56)	No significant association
Hattori et al. 2013 ²⁰	Japan	No	388	61%	M: 21/79 ^a F: 62/38 ^a	2 national surveys M: 32/68 F: 76/24	NA	With the exception of men in their 30s, the prevalence was greater in all age groups compared with the general population.
Demirkok et al. 2007 ⁸⁰	Turkish	No	267	69%	T: 71/29 M: 46/54 F: 83/17	National survey M smokers: 52–65% F smokers: 38–49%	NA	Smokers less frequent in F patients with sarcoidosis than those in normal population.
Musellim et al. 2009 ¹⁰	Turkish	No	293	68%	T: 73/6/ 21 M: 48/52 ^a F: 85/15 ^a	National survey M smokers: 65% F smokers: 25%	NA	Smokers less frequent in both F and M patients with

References	Country	Case-control study	Number of patients with smoking history Sarcoidosis/controls	Female	NS/S or NS/FS/S or NS/ES ^a M/F or Total (%)	Control group	Adjusted Odds Ratio (95% CI) or Relative Risk (95% CI) ^b	Association between smoking and sarcoidosis
Haraldsdóttir et al. 2021 ⁹	Iceland	No	322	50%	M: 39/41/20 F: 52/33/15	National survey	NA	sarcoidosis than those in normal population.
Extra-pulmonary sarcoidosis								
Cardiac sarcoidosis								
Xu et al. 2022 ²⁹	Canada	Yes	82/820	45%	72/28 ^a	52/48 ^a	0.94 (0.89–0.99)	No significant difference of sarcoidosis incidence between smoking habits NS/ FS/S (incidence was found by comparing the study group with national data).
Spence et al. 2020 ³⁰	Canada	Yes	43/36		81/19 ^a	53/47 ^a	0.14 (0.04–0.51)	Negative association
Ocular sarcoidosis								
Janot et al. 2015 ³¹	USA	No	109 (21 with OS/88 without OS)	43% with OS 74% without OS	With OS: 29/71 ^a	Without OS: 58/42 ^a	5.24 (1.58–17.41)	Positive association
Sarcoid arthritis								
Visser et al. 2002 ³³	Netherlands	No	55	49%	93/7	National survey (12,827) 65/35	0.14 (0.05–0.39)	Negative association

CI: confidence interval; ES: ever-smokers; F: female; FS: former smokers; M: male; NA: non available; NS: non-smokers; OS: ocular sarcoidosis; S: smokers; SMS: swedish moist snuff; T: total.

^aNS/ES (Non-smokers/Ever-smokers).

^bRelative Risk (95% CI).

^cUnadjusted odds ratio.

Table 2.

Summary of bronchoalveolar lavage studies in patients with pulmonary sarcoidosis.

References	Number of patients	NS/S or NS/ES or NS/FS/S	Comparison groups	Total cell count Mean $\pm$ SD or mean $\pm$ SEM ^a	Alveolar macrophages (%) Mean $\pm$ SD or mean $\pm$ SEM ^a	Lymphocytes Mean $\pm$ SD or mean $\pm$ SEM ^a ; median (range)	Tlymphocytes Mean $\pm$ SD or mean $\pm$ SEM ^a	Neutrophils Mean $\pm$ SD or mean $\pm$ SEM ^a	Cytokines Mean $\pm$ SD
<i>b</i> ₁ Dauber et al. 1979 ⁶³	14	NA	NS vs. S	NA	NA	NA	T cells: 4.19 $\pm$ 0.72 vs. 0.896 $\pm$ 0.107 $\times$ 10 ⁴ cells/ml (<i>p</i><0.001) Activated T cells: 1.04 $\pm$ 0.478 vs. 0.148 $\pm$ 0.032 $\times$ 10 ⁴ cells/ml (<i>p</i><0.01)	NA	NA
<i>b</i> ₁ Leuenberger et al. 1985 ⁶¹	NA	NA	NS vs. S	NS<S (<i>p</i><0.02)	60% vs. 83% (<i>p</i><0.002)	Absolute number: ns Percentage: 14% vs. 34% (<i>p</i><0.002)	NA	NA	NA
Lawrence et al. 1986 ⁶⁶	10	4/6	NS vs. S	NA	NA	NA	CD4: 56% vs. 62% (ns) CD8: 13% vs. 16% (ns) Ratio CD4/CD8: 4.3 vs. 4.6 (ns)	NA	NA
Valeyre et al. 1988 ²⁴	64	45/19	NS vs. S	28.0 $\pm$ 1.3 vs. 39.4 $\pm$ 18.4 $\times$ 10 ⁴ cells/ml (<i>p</i><0.05)	59 $\pm$ 17% vs. 68 $\pm$ 14% (ns) <i>Note: S>20 cigarettes/d had more alveolar macrophages than NS (p<0.05).</i>	39 $\pm$ 17% vs. 30 $\pm$ 14% (<i>p</i><0.05)	CD4: 78.9 $\pm$ 14% vs. 65.8 $\pm$ 15.7% (<i>p</i><0.01) CD8: 15.4 $\pm$ 10% vs. 30.1 $\pm$ 14.8% (<i>p</i><0.001) Ratio CD4/CD8: 8.1 $\pm$ 5.9 vs. 3.2 $\pm$ 2.7 (<i>p</i><0.01)	1.4 $\pm$ 1.3% vs. 1.3 $\pm$ 0.8% (ns)	NA
Yamaguchi et al. 1989 ³⁵	40	22/18	NS vs. S	19 $\pm$ 2 vs. 31 $\pm$ 4 $\times$ 10 ⁴ cells/ml (<i>p</i><0.01) ^a	58 $\pm$ 4% vs. 85 $\pm$ 3% (<i>p</i><0.0001) ^a	41 $\pm$ 4% vs. 14 $\pm$ 3% (<i>p</i><0.0001) ^a 8.2 $\pm$ 14 vs. 4.1 $\pm$ 9 $\times$ 10 ⁴ cells/ml (<i>p</i><0.05) ^a	Ratio CD4/CD8: 7.0 $\pm$ 2.6 vs. 4.7 $\pm$ 1.1 (<i>p</i><0.001) ^a	0 $\pm$ 0% vs. 1 $\pm$ 0% (ns) ^a	IL-1/IL-1 β released from LPS-stimulated alveolar macrophages: significantly decreased in S compared with NS (<i>p</i><0.05/ <i>p</i><0.01)
Yamaguchi et al. 1993 ⁵⁶	29	13/16	NS vs. S	20 $\pm$ 10 vs. 31 $\pm$ 16 $\times$ 10 ⁴ cells/ml (<i>p</i><0.0001)	60 $\pm$ 22% vs. 85 $\pm$ 13% (<i>p</i><0.001)	40 $\pm$ 23% vs. 15 $\pm$ 13% (<i>p</i><0.001)	NA	0.3 $\pm$ 0.5% vs. 0.3 $\pm$ 0.4% (ns)	TNF released from LPS-stimulated alveolar macrophages: significantly decreased in S compared with NS (322 $\pm$ 160 vs. 570 $\pm$ 329U/ml, <i>p</i><0.01)
Pueringer et al. 1993 ⁶⁷	68	48/16/4	NS vs. S	NA	NA	NA	NA	NA	TNF (LPS stimulated), IL-1,

References	Number of patients	NS/S or NS/ES or NS/FS/S	Comparison groups	Total cell count Mean $\pm$ SD or mean $\pm$ SEM ^a	Alveolar macrophages (%) Mean $\pm$ SD or mean $\pm$ SEM ^a	Lymphocytes Mean $\pm$ SD or mean $\pm$ SEM ^a , median (range)	Tlymphocytes Mean $\pm$ SD or mean $\pm$ SEM ^a	Neutrophils Mean $\pm$ SD or mean $\pm$ SEM ^a	Cytokines Mean $\pm$ SD
Drent et al. 1993 ⁵⁷	100 (18 A, 60 B, 22 C)	A: 11/7 B: 50/10 C: 16/6	NS vs. S	A: 19.2 $\pm$ 3.5 vs. 31 $\pm$ 5.5 $\times$ 10 ⁶ cells B: 20.8 $\pm$ 1.8 vs. 43.9 $\pm$ 14.9 $\times$ 10 ⁶ cells C: 23.5 $\pm$ 3.2 vs. 31.2 $\pm$ 7.0 $\times$ 10 ⁶ cells	NA	A: 31.3 $\pm$ 5.1% vs. 21.3 $\pm$ 3.6% (<i>p</i> < 0.001) ^a B: 38.1 $\pm$ 2.6% vs. 27.1 $\pm$ 6.4% (<i>p</i> < 0.001) ^a C: 40.2 $\pm$ 4.5% vs. 18.7 $\pm$ 5.9% (<i>p</i> < 0.001) ^a	A: CD4% and CD8% (ns); CD4/CD8 ratio 4.7 $\pm$ 1.1 vs. 5.6 $\pm$ 1.6 (ns) ^a B: CD4% and CD8% (ns); CD4/CD8 ratio 8.0 $\pm$ 1.2 vs. 5.6 $\pm$ 1.2 (<i>p</i> < 0.05) ^a C: CD4% and CD8% (ns); CD3+ LT 88.8 $\pm$ 2.1% vs. 95.3 $\pm$ 1.2% (<i>p</i> < 0.05) ^a ; CD4/CD8 ratio 10.7 $\pm$ 1.5 vs. 8.3 $\pm$ 1.8 (<i>p</i> < 0.05) ^a	NA	and PGE2 release from alveolar macrophages was not consistently related to the smoking status.
Suzuki et al. 1996 ⁵⁸	166	96/70	NS vs. S	NS < S (<i>p</i> < 0.01)	62.2 $\pm$ 2.0% vs. 74.3 $\pm$ 2.1% (<i>p</i> < 0.01)	35.7 $\pm$ 2.0% vs. 24.8 $\pm$ 2.1% (<i>p</i> < 0.01)	The BALF Ia+ Tlymph (CD3+ HLA-DR+) % value was not influenced by smoking.	NA	NA
Shigemura et al. 2010 ⁵⁹	64	30/11/22	NS vs. S (FS excluded)	NS < S (<i>p</i> < 0.01)	NA	44% (11.7–76.3) vs. 17.1% (0.3–73.7) (<i>p</i> < 0.05) Absolute number: ns	CD4 absolute number: ns	NA	NA
Urbankowski et al. 2012 ⁶⁰	61	35/26	NS vs. ES	14.4 $\pm$ 7.7 vs. 21.1 $\pm$ 14.7 $\times$ 10 ⁶ cells	7.7 $\pm$ 4.7 vs. 12.5 $\pm$ 10.8 $\times$ 10 ⁶ (<i>p</i> < 0.05) 52 $\pm$ 17% vs. 64 $\pm$ 23% (<i>p</i> < 0.05)	39.2 $\pm$ 18.5% vs. 25.6 $\pm$ 19.5% (<i>p</i> < 0.05)	NA	6.8 $\pm$ 3.9% vs. 6.8 $\pm$ 9.0% (ns)	Median concentration of IL-6, IFN- γ , and IL-17A: no significant differences between NS and S
Osinska et al. 2014 ⁶²	36	18/18	NS vs. S	NA	8.7 vs. 15.2 $\times$ 10 ⁶ cells/ml (<i>p</i> = 0.03)	NA	NA	NA	NA
Aleksionienė et al. 2017 ⁶⁴	80	64/16	NS vs. S	NA	NA	NA	CD4: 68% vs. 53% (<i>p</i> = 0.035) CD8: 22% vs. 40% (<i>p</i> = 0.005) CD4/CD8 ratio: 5.1 vs. 2.8 (<i>p</i> = 0.011) Note: these data are for Stage II (not enough smokers in stage I and III)	NA	NA
Akyıl et al. 2020 ⁶⁵	257	192/65	NS vs. ES	NA	40.1% vs. 44.9% (<i>p</i> = 0.087)	42% vs. 39.3% (<i>p</i> = 0.353)	CD4/CD8 ratio: 5.66 vs. 5.68 (<i>p</i> = 0.989)	16.9% vs. 13.7% (<i>p</i> = 0.044)	NA

BALF: bronchoalveolar lavage fluid; ES: ever-smokers; FS: former smokers; IQR: interquartile range; LPS: lipopolysaccharide; NA: not available; NS: non-smokers; ns: non-significant; S: smokers; SD: standard deviation; SEM: standard error of mean.

^aMean ± SEM.
^bAbstract.

Table 3.

Summary of pulmonary function test studies in patients with pulmonary sarcoidosis.

References	Number of patients	NS/S or NS/ES or NS/FS/S	Comparison groups	Total Lung Capacity (% pred)	FEV ₁ (% pred) mean $\pm$ SD or median (IQR)	FVC (% pred) mean $\pm$ SD or median (IQR)	FEV ₁ /FVC ratio (%) mean $\pm$ SD or SEM ^a ; median (IQR)	DL _{CO} (% pred or ml/min/mm Hg ^b) mean $\pm$ SD or SEM ^a ; median (IQR)	Comments
Valeyre et al. 1988 ²⁴	64	42/16	NS vs. S	88 $\pm$ 13 vs. 88 $\pm$ 17 (ns)	NA	NA	97 $\pm$ 8 vs. 94 $\pm$ 9 (ns)	74 $\pm$ 17 vs. 74 $\pm$ 14 (ns)	Both groups had a similar percentage of patients with <i>a</i> > 20% decrease from predicted values.
Yamaguchi et al. 1989 ⁵⁵	40	22/18	NS vs. S	NA	NA	NA	83 $\pm$ 1 vs. 85 $\pm$ 1 (ns) ^a	107 $\pm$ 4 vs. 89 $\pm$ 2 (<i>p</i> < 0.001) ^a	NA
Hanley et al. 1991 ⁷³	30	17/13	NS vs. S	82.5 $\pm$ 3.3 vs. 72.7 $\pm$ 3.5 (<i>p</i> < 0.05)	NA	76.2 $\pm$ 4.1 vs. 65.8 $\pm$ 2.8 (<i>p</i> = 0.06)	79.8 $\pm$ 2.5 vs. 79.9 $\pm$ 3.4 (ns)	111 $\pm$ 5.7 vs. 111.3 $\pm$ 6.8 (ns)	NA
Terasaki et al. 2005 ⁷¹	46	23/23	NS vs. S	NA	101.9 $\pm$ 18.6 vs. 97.0 $\pm$ 21.5 (<i>p</i> = 0.404)	111.4 $\pm$ 15.5 vs. 107.4 $\pm$ 17.9 (<i>p</i> = 0.415)	77.2 $\pm$ 9.6 vs. 77.2 $\pm$ 12.5 (<i>p</i> = 0.992)	NA	Spearman's correlation indicated that air trapping negatively correlated with FVC in S (<i>p</i> < 0.05), but not in NS.
Gerke et al. 2011 ⁷⁰	124	86/18/20	NS vs. ES	89 $\pm$ 17 vs. 88 $\pm$ 19 (<i>p</i> = 0.7)	83 $\pm$ 24 vs. 83 $\pm$ 22 (<i>p</i> = 0.98)	86 $\pm$ 18 vs. 85 $\pm$ 21 (<i>p</i> = 0.8)	74 (66–78) vs. 77 (70–82) (<i>p</i> = 0.08)	76 (66–88) vs. 71 (55–83) (<i>p</i> = 0.1)	NA
^c Sahoo et al. 2012 ⁷²	693	316/377 (baseline) 105/136 (2-year FU)	NS vs. ES	NA	88 (76–98) vs. 83 (71–95) (<i>p</i> = 0.002)	89 (79–99) vs. 87.5 (76–97) (<i>p</i> = 0.043)	NA	NA	2-year changes in PFTs, dyspnea, and therapy needs were not significantly linked to smoking status.
Urbankowski et al. 2012 ⁶⁰	61	35/26	NS vs. ES	106.0 $\pm$ 15.3 vs. 102.0 $\pm$ 7.0 (ns)	103.9 $\pm$ 21.9 vs. 90.5 $\pm$ 16.4 (<i>p</i> < 0.05)	117.1 $\pm$ 27.8 vs. 98.1 $\pm$ 15.9 (<i>p</i> < 0.05)	NA	26.1 $\pm$ 7.5 vs. 26.5 $\pm$ 7.5 (ns) ^c	NA
Krell et al. 2012 ³²	518	360/158	NS vs. S	79 $\pm$ 17 vs. 81 $\pm$ 18 (<i>p</i> = 0.3)	73.4 $\pm$ 17 vs. 69.5 $\pm$ 19 (<i>p</i> = 0.04)	74.8 $\pm$ 18 vs. 72 $\pm$ 19 (<i>p</i> = 0.16)	77.5 $\pm$ 10 vs. 74 $\pm$ 11 (<i>p</i> = 0.01)	65.5 $\pm$ 20.7 vs. 53.8 $\pm$ 22.6 (<i>p</i> < 0.001)	Effects are more pronounced in female smokers with sarcoidosis than in males.
Domagala-Kulawik et al. 2013 ⁶⁸	57	36/14/7	NS vs. ES	NA	105.6 $\pm$ 15.4 vs. 90.7 $\pm$ 16.8 (<i>p</i> < 0.001)	117.1 $\pm$ 25 vs. 98.3 $\pm$ 16.1 (<i>p</i> < 0.001)	NA	NA	NA

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DLCO: diffusing capacity of the lungs for carbon monoxide; ES: ever-smokers; FEV₁: forced expiratory volume in one second; FS: former smokers; FU: follow up; FVC: forced vital capacity; IQR: interquartile range; NA: not available; NS: non-smokers; ns: non-significant; PFTs: pulmonary function tests; S: smokers; SD: standard deviation; SEM: standard error of mean.

^aMean ± SEM.

^bDLCO ml/min/mm Hg.

^cAbstract.